

Response to second reviewer's comments

Concerning paper “Generalised early warning signals in multivariate and gridded data with an application to tropical cyclones”

by *J. Prettyman, T. Kuna, V. N. Livina*

We thank the reviewer for the time taken to read through this submission. We regret that the reviewer has previously had to review the manuscript for another magazine — being called by different journals to review the paper only confirms the reputation of the reviewer as an expert in this area. We note that although our previous major revision did address most the reviewer's comments, the reservation of the reviewer regarding the choice of the case study has not been addressed, although we have been able to improve the manuscript significantly.

The motivation of the use of the tropical cyclones example was based on the observation that the sudden drop in pressure does represent an example of a tipping point in a geophysical system. Indeed, a tipping point is not required to be bifurcational, as noted by [Ashwin et al. Royal Soc Proc A, 2012] who describe noise-induced and rate-induced tipping. Although the tipping impetus is external to the system in this case, we have still classified this as a tipping point. Moreover, the dynamics of the atmospheric pressure during a cyclone is more complex than simple decline (trend), as can be seen in the PS-indicator, which detects the change in spectral properties of the data.

As noted by the reviewer in that previous review, many readers of that magazine would be interested in the practical prediction problem - particularly of tropical cyclones. Indeed, this work is not of immediate practical impact in this context, and that is why we chose to resubmit to a different journal where the focus might be less on the application to meteorology and more on the methods introduced.

The aim of the paper, insofar as we are concerned with the tropical cyclones example, is to apply a tipping point indicator, shown to work well in a bifurcating dynamical system such as the Van der Pol oscillator, to a meteorological example of a non-bifurcational tipping point. In this case, the early warning signal does not manifest at the 1σ significance level, but in future work we would like to apply these methods to other physical examples of non-bifurcational tipping, if suitable examples can be found.

We would like to stress that the real-world examples of multi-dimensional tipping points, unlike one-dimensional trajectories that have been studied so far in multiple publications, are very limited.

We were particularly motivated by the reviewer's suggestion to investigate the bistability of the Sahara and have been in conversation with Max Plank Institute colleagues

to gain rainfall and pollen data from the ECHAM6-HAM2 model, which can be used as a proxy for Saharan greening. Unfortunately, these data are not yet ready for use and not available in public domain. Previous models used in the same context have given snapshots at 100-year intervals, which is unsuitable for the methods presented here. We very much hope that the greening of the Sahara will be a subject of our future research.

For now, we believe there is merit in applying tipping point analysis to systems exhibiting non-bifurcational tipping, in order to expand this area of research. We also believe there is merit, generally, in the methods presented in this submission, aside from the question of the particular meteorological example used. Our approach is novel, it has been introduced at the moment of rising interest in multi-dimensional tipping points, and it certainly can be later improved and generalised by other researchers (in particular, using other techniques, as the reviewer suggested when mentioned that they could reproduce our results with an alternative approach). At this stage of development, in our opinion, our results need to be published, even if imperfect and subject to further improvement.

We thank the reviewer again for their time in responding to this submission. We hope that the revised manuscript is now suitable for publication in the Chaos magazine.

Yours faithfully,

J. Prettyman, T. Kuna, V. N. Livina